

Problem 4

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Problem 1 The **velocity** function for a man walking along a straight road which runs east and west is given by $v(t) = -t^2 + 4t - 3$ ft/min.

- (a) Set up a definite integral for the man's **displacement** during the time interval from 2 minutes to 6 minutes after he began running.
- (b) **At home:** Evaluate the definite integral using the limit of a right Riemann sum.
- (c) Is this the same as the total **distance** the man walked from 2 minutes to 6 minutes? Why or why not?